Continuous True-Time Delay Phase Shifter Using Distributed Inductive and Capacitive Miller Effect

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***Abstract*— A new true-time delay phase shifter concept is proposed exploiting the distributed Miller effect in coupled transmission lines. Simultaneous change in Miller capacitance and inductance controlled by a single analog voltage changes the propagation delay of a transmission line with constant input impedance and insertion loss. The true-time delay line feature of the proposed architecture accommodates a wide bandwidth signal without group delay distortion. A theoretical framework for understanding Miller inductance from voltage and current duality and the distributed Miller effect using coupled wave equations is presented. As a proof of concept, a four-stage tunable delay line with a high-speed phase modulation capability was fabricated in a 45-nm RF SOI CMOS process and occupies an active area of 0.28 mm^2. The measured IC demonstrates a broadband tunable true-time delay with a measured group delay tuning range of 18 ± 1 ps from 11 to 24 GHz. The measured phase shift ranges are 173° and 182° with insertion losses of 10.6 ± 0.7 dB and 11.6 ± 1 dB at 28 and 30 GHz, respectively. The measured dc power consumption is 22 mW from a 1.2-V supply.**

***Index Terms*— Ka-band, Miller effect, phase modulation, phase shifter, phased array, true-time delay, 28 GHz, wideband.**

I. Introduction

A WIDE-BANDWIDTH phase shifter is an essential component in millimeter-wave (mmWave) phased arrays to perform beam forming and steering in the applications of next-generation communication (5G) and high-resolution radar systems [1]–[3]. When large fractional bandwidth signals are processed in a phased array as shown in Fig. 1(a), each phase-shifting front-end element ideally compensates for the time difference over signal bandwidth given by an integer multiple of τ_0, where $\tau_0 = \Delta x/c$, $\Delta x = d\sin(\theta_s)$, d is the element spacing, θ_s is a scan angle, and c is the velocity of the electromagnetic (EM) wave. However, commonly used mmWave phase shifters, such as reflection-type phase shifters [4]–[6], switched *LC*-type phase shifters [7], [8], and vector modulator-based phase shifters [9]–[11], generally aim to compensate for a phase shift given by an integer multiple of $\phi_0 = 2\pi f_c \tau_0$ around the carrier frequency of f_c within a narrow bandwidth. The use of such phase shifters for a wideband signal results in delay variation within the signal bandwidth, which broadens the beamwidth and reduces array resolution as beam pointing is different for different frequency components [12], [13]. In addition, when pulses are transmitted in a phased array using the narrowband phase shifters, the propagation time difference between the first and last elements, $N\tau_0$, should be much smaller than the pulsewidth τ_{pulse} to combine pulses coherently in time with beam forming, where N is the element number [14]. This quasi-monochromatic approximation is written by

$$N\tau_0 = \frac{N\Delta x}{c} \ll \tau_{\text{pulse}} \cong \frac{1}{BW} \tag{1}$$

which indicates that the signal bandwidth BW is limited by the array system parameters $c/(N\Delta x)$ to avoid a signal distortion. To overcome those limitations, true-time delay phase shifters can replace the narrow-bandwidth phase shifters to implement

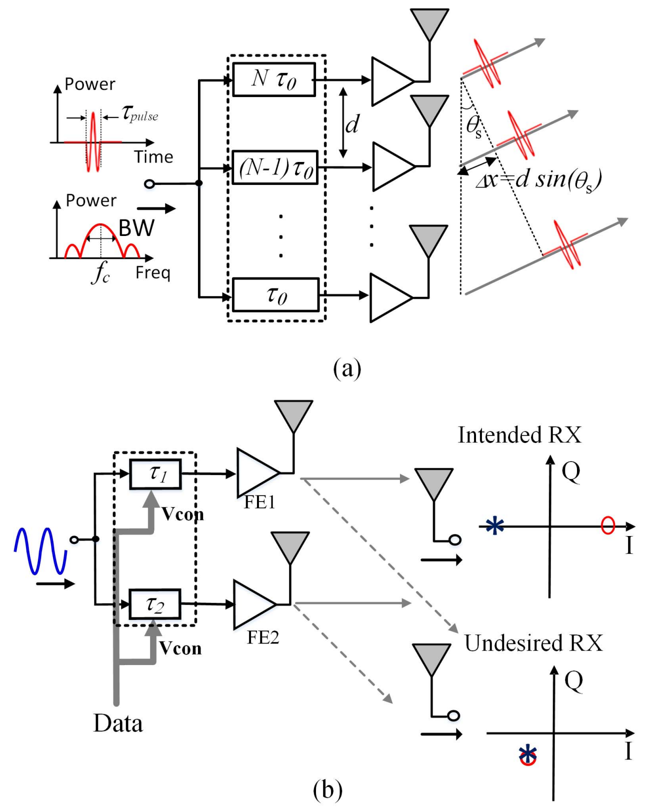

Fig. 1. Applications of the proposed tunable delay line in (a) ultra-wideband timed arrays and (b) directional modulation for improved physical layer security.

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a timed array, the time-domain counterpart of a phased array, to enable high data rates in communications and fine range resolution in radars [15], [16]. The true-time delay phase shifter, or simply a tunable delay line, intrinsically exhibits a constant group delay shift across frequency for a wide signal bandwidth. In addition to the true-time delay feature, the following features are desirable for tunable delay lines from a system performance perspective: 1) high time resolution, which leads to fine beam steering without compromising sidelobe rejection [1]; 2) fast delay switching speed linked to user switching speed in spatial multiplexing communication and target scanning speed in radar; and 3) phase-invariant insertion loss and linear phase control enabling simplified amplitude and phase calibration of front-end circuitry.

In this paper, we propose a continuously tunable delay line based on the distributed Miller effect in coupled transmission lines to achieve these features. As a proof of concept, a four-stage delay line is implemented in a 45-nm RF SOI CMOS process to demonstrate a wideband continuously tunable delay, achieving $>180°$ phase tuning range at 30 GHz with an insertion loss variation < 1 dB. The proposed design is also capable of modulating the phase with a single analog voltage at a high speed. This feature can be potentially used for directional modulation in mmWave phased arrays, which creates the desired phase in the desired direction for each symbol in a data transmission, while scrambling the symbols in other undesired directions for improved physical layer security as shown in Fig. 1(b) [17], [18].

This paper is an expanded version of a conference publication [19]. Although the conference publication gave a brief introduction to the theory and described the IC implementation and key measurement results, this paper provides: 1) review of prior work; 2) extended theory, including discussion of the amplifier time delay effect; 3) design implementation and simulation details of the variable gain amplifier (VGA) and coupled transmission lines; and 4) comprehensive measurement results including a group delay measurement, a measurement with wideband modulated signals, a large signal measurement, data at various temperatures and from multiple samples, and detailed comparison to simulations. Section II reviews prior work in wideband tunable delay lines. Section III reviews the theory of the distributed Miller effect. Section IV describes circuit-level implementation. Section V shows on-wafer measurement results, and Section VI provides conclusions.

II. Prior Work on Tunable Delay Lines

A wide bandwidth time delay can be simply implemented with a continuous transmission line since it exhibits a constant propagation delay across frequency given by $\tau_{\text{pd}} = d(LC)^{1/2}$, where d is the length of the transmission line and L and C are the inductance and capacitance per unit length, respectively [20]. Hence, τ_{pd} can be varied by changing d, L, or C, which leads to the three types of tunable transmission line designs as illustrated in Fig. 2. The propagation length of the line can be switched in a trombone-line structure by turning one amplifier at a time [15] as shown in Fig. 2(a): when an amplifier farther

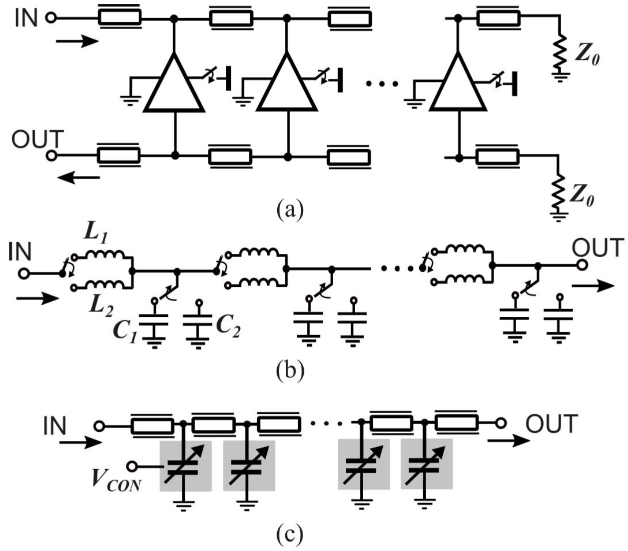


Fig. 2. Prior work on tunable transmission lines. (a) Trombone-line architecture. (b) Transmission line with switched inductance and capacitance. (c) Artificial transmission line periodically loaded with analog varactors.

from the input and output ports is selected, a longer delay is generated. As shown in Fig. 2(b), the propagation delay can be also tuned by switching only capacitance [21], [22], only inductance [23], or both inductance and capacitance [24], [25]. When both inductance and capacitance change in the same direction and ratio, both a broader delay tuning range and also a constant characteristic impedance can be achieved while effecting delay control. However, the resolution of such switch-based architectures with segmentation is limited by the minimum size of the switches and passives that can be reliably implemented for a given fabrication process. For finer resolution, analog varactors such as accumulation-mode MOS varactors in a CMOS process can be used to periodically load a transmission line, yielding a continuously tunable delay as shown in Fig. 2(c) [26]. However, the characteristics of accumulation-mode MOS varactors, which exhibit a reduced quality factor (Q) at higher frequency, limit the use of this approach in mmWave phase shifters. More importantly, a strong dependence of Q on control voltage can result in a large and highly undesirable loss variation over phase tuning [27], [28].

III. Theory on Distributed Miller Effect

A. Miller Effect and Prediction of Miller Inductance

According to the Miller effect [29], an input current I_{IN} flowing into the feedback impedance Z_F of an inverting voltage amplifier with a gain of A_V is boosted by a factor of $A_V + 1$ for given input voltage V_{IN}. In other words, the feedback impedance Z_F seen at the input is reduced by the same factor, which is given by

$$Z_{\text{in}} = Z_F/(A_V + 1). \tag{2}$$

When Z_F is capacitive, the input capacitance contributed by Z_F through the Miller effect, or simply Miller capacitance C_M, is given by $C_M = (A_V+1)C_c$, by plugging $Z_F = (1/sC_c)$ and

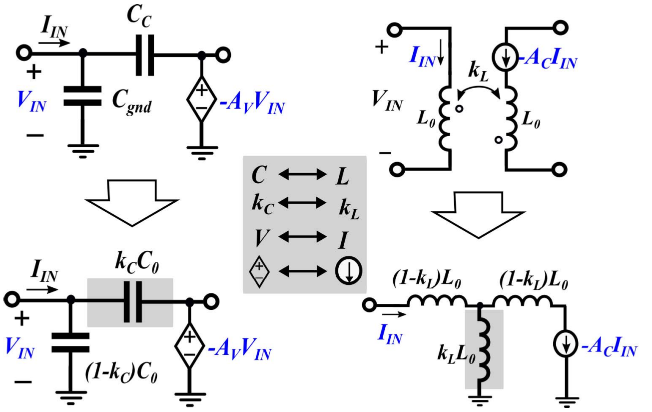


Fig. 3. Generalization of Miller capacitance by including input-to-ground capacitance and prediction of Miller inductance exploiting the duality, where $C_0 = C_{\text{gnd}} + C_c$ and $k_c = C_c/C_0$.

$Z_{\text{in}} = (1/sC_M)$ into (2). By generalizing this result, the input-to-ground capacitance C_{gnd} can be absorbed into the Miller capacitance by introducing a capacitive coupling factor, k_c, which represents a portion of feedback (coupling) capacitance C_c of the total capacitance $C_0 = C_c + C_{\text{gnd}}$ as shown in Fig. 3 (left). Then, the Miller capacitance can be written as $C_M = C_{\text{gnd}} + (1 + A_V)C_c = C_0(1 + k_c A_V)$. Exploiting the duality between voltage and current for capacitance and inductance, the existence of a Miller inductance, a boosted coupling (mutual) inductance seen at the input through a current amplifier, can be predicted as shown in Fig. 3 (right). The Miller inductance L_M is written by $L_M = L_0(1+k_L A_C)$, the dual form of the Miller capacitance C_M. As a result, both capacitance and inductance seen at the input C_M and L_M are tunable with voltage and current gains through the Miller effect, respectively.

B. Tunable Delay Line Through Distributed Miller Effect

Next, a tunable transmission line can be composed of Miller capacitance and inductance to simultaneously change inductance and capacitance on the signal line as shown in Fig. 4, where a VGA provides both voltage and current amplification on the amplifier line, and where L_0 and $C_0 = C_c + C_{\text{gnd}}$ are the inductance and total capacitance per unit length, respectively. It is noted that both voltage and current gain, A_V and A_C, are given by $A_V = A_C = A = G_m Z_0$, where G_m is the transconductance and Z_0 is the characteristic impedance of the amplifier line. In a short length of transmission lines where power exchange between the two lines can be ignored, the propagation delay per unit length (τ_{pdu}) and the characteristic impedance (Z_0) seen by the input port of the signal line are given by

$$\tau_{\text{pdu}} = \sqrt{L_M C_M} = \sqrt{(1+k_L A)(1+k_c A)} \cdot \sqrt{L_0 C_0} \quad (3a)$$

$$Z_0 = \sqrt{\frac{L_M}{C_M}} = \sqrt{\frac{1+k_L A}{1+k_c A}} \cdot \sqrt{\frac{L_0}{C_0}}. \quad (3b)$$

When capacitive coupling factor k_c and inductive coupling factor k_L are equal to k, (3) can be simplified to $\tau_{\text{pdu}} = (1 + kA)(L_0 C_0)^{1/2}$ and $Z_0 = (L_0/C_0)^{1/2}$, which indicates that the propagation delay is amplified by a factor of $1 + kA$, whereas the characteristic impedance seen by the input port is invariant with the gain, which is desirable for tunable delay lines. The result of this analysis is that the propagation delay is linearly related to the amplifier gain A. However, as the length of the transmission lines becomes longer, the power exchange between the two lines through coupling is not negligible during propagation, affecting the Miller effect.

To achieve a more complete understanding, coupled wave equations between signal and amplifier lines can be solved by conceptually decomposing waves into common and differential modes as shown in Fig. 4(b). The signal and amplifier lines are driven by V_0 and $-AV_0$ which can be decomposed into

$$\begin{bmatrix} 1 \\ -A \end{bmatrix} V_0 = \begin{bmatrix} 1 & 1/2 \\ 1 & -1/2 \end{bmatrix} \begin{bmatrix} V_{0c} \\ V_{0d} \end{bmatrix} \quad (4)$$

where $V_{0c} = (1 - A)V_0/2$ and $V_{0d} = (1 + A)V_0$. As shown in Fig. 4(b), the inductance is reduced by a factor $1 - k_L$ for the common mode since the common mode currents flow in the same direction and the inductors are coupled in opposite directions. The equivalent capacitance is only C_{gnd}, without contribution from the coupling capacitor C_c. After C_{gnd} is rewritten as $C_{\text{gnd}} = (1 - k_c)C_0$, the propagation constant β_c and characteristic impedance Z_{0c} can be expressed as

$$\beta_c = \omega\sqrt{(1-k_L)(1-k_c)} \cdot \sqrt{L_0 C_0} \quad (5a)$$

$$Z_{0c} = \sqrt{(1-k_L)/(1-k_c)} \cdot \sqrt{L_0/C_0}. \quad (5b)$$

For the differential mode, the inductance is increased by a factor of $1+k_L$, and the net capacitance is $C_{\text{gnd}}+2C_c$, which can be rewritten as $(1 + k_c)C_0$. The propagation constant β_d and the characteristic impedance Z_{0d} are

$$\beta_d = \omega\sqrt{(1+k_L)(1+k_c)} \cdot \sqrt{L_0 C_0} \quad (6a)$$

$$Z_{0d} = \sqrt{(1+k_L)/(1+k_c)} \cdot \sqrt{L_0/C_0}. \quad (6b)$$

Z_{0c} and Z_{0d} are equal to $(L_0/C_0)^{1/2}$ when $k_L = k_C = k$, and the voltages on the signal and amplifier lines at the propagation distance of x, $V_s(x)$ and $V_a(x)$, are the superposition of common and differential modes with different propagation constants, which can be written as

$$\begin{bmatrix} V_s(x) \\ V_a(x) \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1 & -1/2 \end{bmatrix} \begin{bmatrix} V_{0c} \cdot \exp(-\gamma_c x) \\ V_{0d} \cdot \exp(-\gamma_d x) \end{bmatrix} \quad (7)$$

where $\gamma_c = \alpha_c + j\beta_c$ and $\gamma_d = \alpha_d + j\beta_d$. α_c and α_d are the attenuation constants for common and differential modes, respectively. Assuming $\alpha_c \cong \alpha_d \cong \alpha$, the equations for $|V_s(x)|$ and $|V_a(x)|$ can be written by

$$|V_s(x)| = V_0 e^{-\alpha x} \times \sqrt{\frac{1+A^2}{2} + \frac{1-A^2}{2}\cos\Delta\beta x} \quad (8)$$

$$|V_s(x)|^2 + |V_a(x)|^2 = e^{-2\alpha x}(1+A^2)V_0^2 \quad (9)$$

respectively, where $\Delta\beta = \beta_d - \beta_c$. For $A = 0$, (8) becomes

$$|V_s(x)| = V_0 e^{-\alpha x}\left|\cos\frac{\Delta\beta x}{2}\right| \quad (10)$$

which implies that the signal is attenuated due to periodic power transfer to the amplifier line with no excitation ($A = 0$)

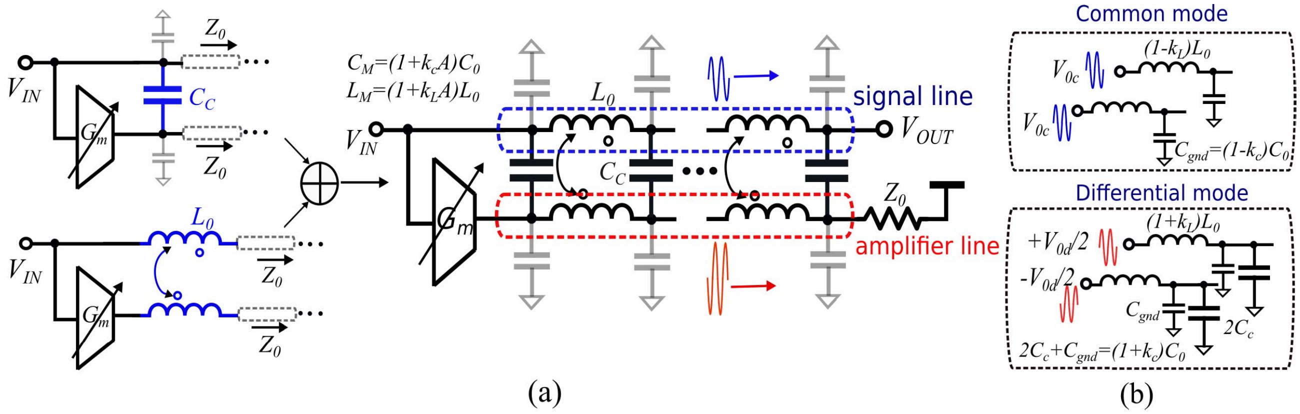


Fig. 4. (a) Distributed Miller capacitance and inductance to form the proposed tunable transmission line and (b) common-mode and differential-mode decomposition.

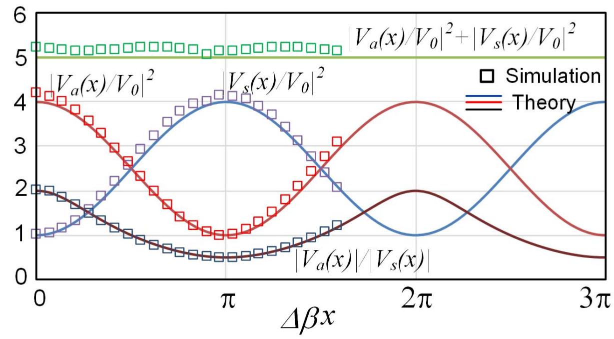


Fig. 5. Periodic power exchange between the signal and amplifier lines and Miller gain $|V_a(x)|/|V_s(x)|$ across propagation distance for amplifier gain of 2 and $\alpha = 0$ compared to the simulated results.

in addition to transmission line loss. Equation (8) indicates that the attenuation becomes lower as A increases, and that once the amplifier line has a higher input power than the signal line ($A > 1$), the signal amplitude $|V_s(x)|$ gradually increases from V_0 at $x = 0$ to its maximum AV_0 at $x = \pi/\Delta\beta$ for $\alpha = 0$. This is a result of power exchange with the amplifier line during propagation, while the total power on the signal and amplifier lines is conserved as stated in (9) for $\alpha = 0$. The maximum power exchange occurs periodically at distances of odd multiples of $\pi/\Delta\beta$. Fig. 5 shows the calculated power exchange compared to the simulated results for $\alpha = 0$. In simulation performed by Cadence (also applied to Figs. 6 and 7), every 50 μm of coupled transmission lines is modeled as a lumped lossless LC segment with L_0' of 20 pH, C_0' of 6.4 fF, and k of 0.55 based on the EM simulation of the implemented coupled lines shown in Fig. 8. Such lumped element modeling is accurate enough at operation frequencies <50 GHz, where the LC cutoff frequency is higher by a factor of >10.

The phase of $V_s(x)$ can be also calculated from (7) as

$$\theta_s(x) = -\arctan\left[\frac{(1-A)\sin\beta_c x + (1+A)\sin\beta_d x}{(1-A)\cos\beta_c x + (1+A)\cos\beta_d x}\right]. \quad (11)$$

Fig. 6 shows the calculated signal gain and phase for different line lengths with respect to amplifier gain based on (8) and (11)

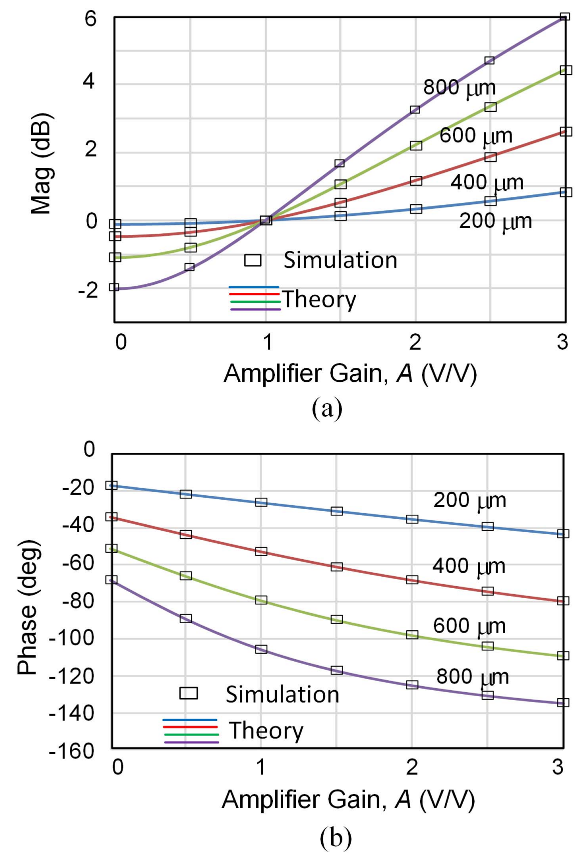


Fig. 6. Calculated (a) magnitude and (b) phase across amplifier gain for different line lengths compared to the simulated results for $\alpha = 0$.

compared to the simulation. For a short length of transmission line, the signal gain does not vary significantly, and the propagation phase shift increases linearly with amplifier gain as $|V_s(x)| \cong V_0$ and $\theta_s(x) \cong -\omega(1 + kA)x(L_0C_0)^{1/2}$ as expected in (3). However, as the line length increases, power exchange between the two lines becomes significant, reducing the Miller gain $|V_a(x)|/|V_s(x)|$ related to the phase shift of $V_s(x)$ as shown in Fig. 5. As a result, the slope of propagation phase shift is reduced with higher gain along with increase in

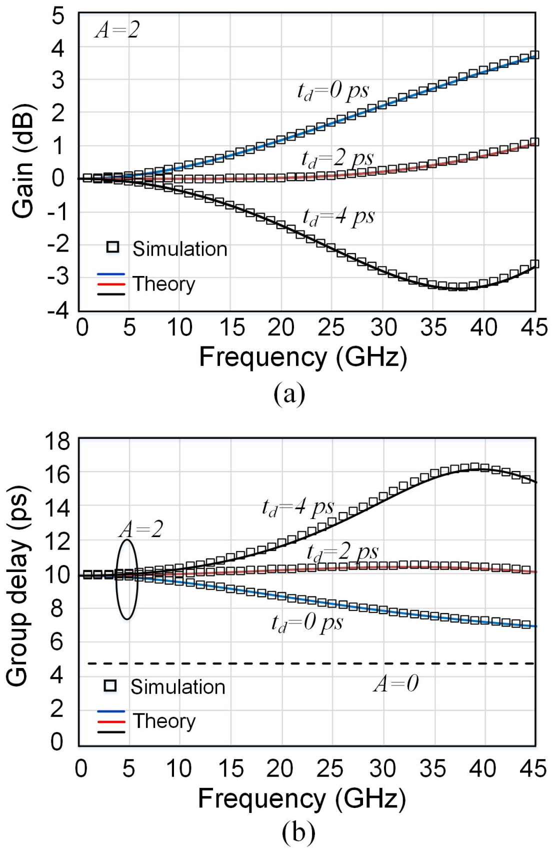


Fig. 7. Calculated (a) gain and (b) group delay across frequency for different amplifier time delays compared to the simulation when x=600 μm and A=2.

signal amplitude. The superposition between the two modes with different propagation phases and amplitudes can be also understood graphically from vector diagrams [19].

C. Effect of Amplifier Time Delay

In the IC implementation, a VGA will exhibit a finite delay time between its input and output. This amplifier delay transforms Miller capacitance to have a positive real part proportional to the amplifier gain for certain phase conditions [30], which increases the signal line loss. By plugging $A = A_0 \exp(-j\omega t_d)$ into (7), where t_d is the amplifier time delay, $|V_s(x)|^2$ and $|V_a(x)|^2$ are written by

$$|V_s(x)|^2 = |V_s(x)|^2_{t_d=0} - (AV_0)^2 \sin \Delta\beta x \sin \omega t_d \quad (12a)$$

$$|V_a(x)|^2 = |V_a(x)|^2_{t_d=0} + (AV_0)^2 \sin \Delta\beta x \sin \omega t_d. \quad (12b)$$

As expected, the signal power is attenuated with the presence of t_d for $\omega t_d < \pi$ as the same amount of power is transferred to the amplifier line. The signal attenuation becomes higher at longer propagation distance until $x = \pi/(2\Delta\beta)$ according to (12). Such opposite power flow compensates for the signal amplitude increase and delay distortion with no amplifier time delay as discussed in Fig. 6. Fig. 7 shows the calculated results of insertion loss and group delay of the signal line across frequency for different amplifier time delays compared to the simulation when the propagation length is 600 μm with the amplifier gain of 2. A longer time delay t_d increases loss and group delay with higher frequency, and $t_d \cong 2$ ps provides a good balance in power flow between the two lines for the minimum insertion loss variation and flat group delay shift over frequency.

IV. IC Implementation in 45-nm RF SOI CMOS

A. Coupled Transmission Line Design

Fig. 8 shows the layout of the implemented coupled transmission lines and associated EM simulation results. The coupled lines are built using the top metal with the ground shield on the lowest level of metal. The signal linewidth is 4 μm and the line spacing is 2 μm. Due to inverted polarity at the amplifier output, the coupling capacitor is formed by side-wall capacitance between the two lines, and the currents of the adjacent two lines are made to flow in the same direction by twisting the lines in the middle of their length to achieve the desired inductive coupling. The coupled lines' dimensions and metal stack were carefully chosen to match characteristic impedances for differential and common modes while including parasitic capacitance from VGAs, which leads to $k_c = k_L$. The length of the coupled line per stage is determined to be 630 μm based on the tradeoff between allowable magnitude variation across delay setting and power consumption. The EM S-parameter model is created using EMX from Integrand Software, Inc. [31]. The phase shift for the differential mode is 95° at 30 GHz, whereas the phase shift for the common mode is 30° with a coupling factor of 0.55 in the EM simulation as shown in Fig. 8(b). The simulated quality factors at 30 GHz are 13 and 6 for differential and common modes, respectively, whereas losses per length are similar (within 0.2 dB) between the two modes as shown in Fig. 8(c).

B. Miller Effect Tunable Delay Line IC Prototype

The prototype of the Miller effect tunable delay line is designed based on the implemented coupled transmission lines as shown in Fig. 9. To achieve >180° tuning at 30 GHz with a small loss variation, four identical stages are cascaded. A differential control signal V_{con} is distributed in an artificial transmission line structure composed of capacitive loadings from VGAs and transmission lines TL_{con} with a 100-Ω termination resistor to support a high-speed phase modulation. To achieve linear gain control with V_{con}, a VGA is implemented with differential cascode pair $M_{2,3}$ as shown in Fig. 9. Higher ΔV_{con} results in more current being steered into the output, increasing the gain. Since the gain can be modulated through the differential pair, the proposed tunable delay line is capable of fast phase modulation. Fig. 10 shows the simulated dc gain of the VGA with respect to ΔV_{con} when the output is terminated with the appropriate characteristic impedance. The group delay across frequency is also simulated with different ΔV_{con}, which ranges from 2.3 to 3 ps. To compensate for the extra amplifier time delay beyond the optimum t_d for the minimum amplitude variation and linear phase control, a transmission line with a length of 230-μm TL_{delay} is placed before the signal line input. The termination impedance of the amplifier line is implemented with a 3-bit variable resistor with

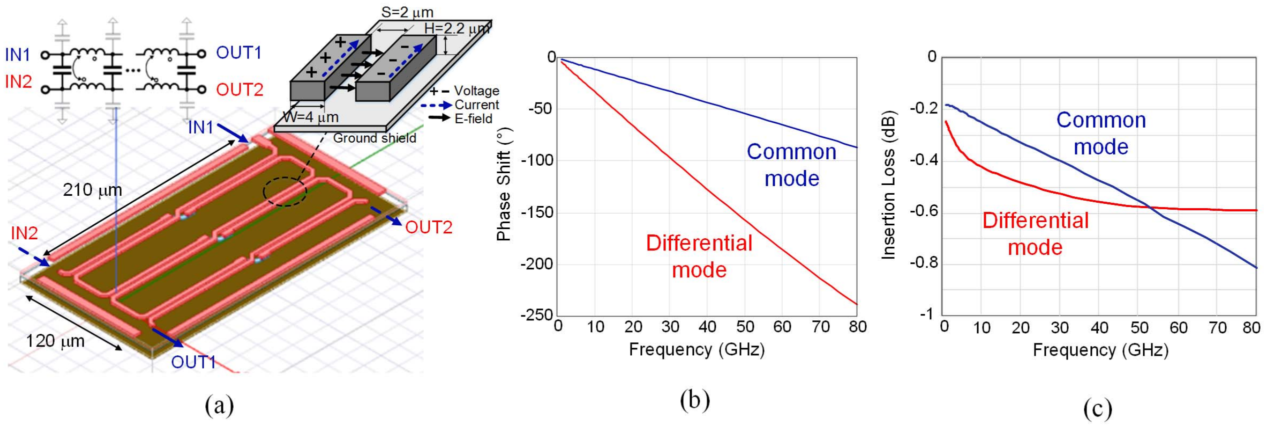


Fig. 8. (a) Coupled transmission line implementation and EM-simulated (b) phase and (c) loss across frequency in common and differential modes.

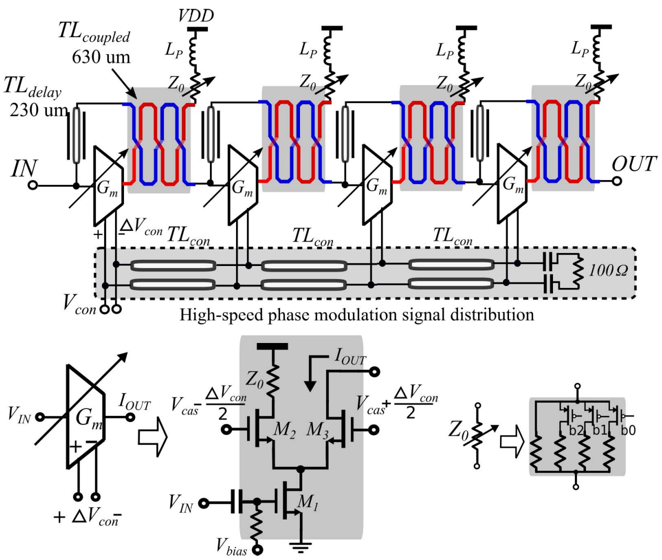


Fig. 9. Implemented four-stage tunable delay line schematic.

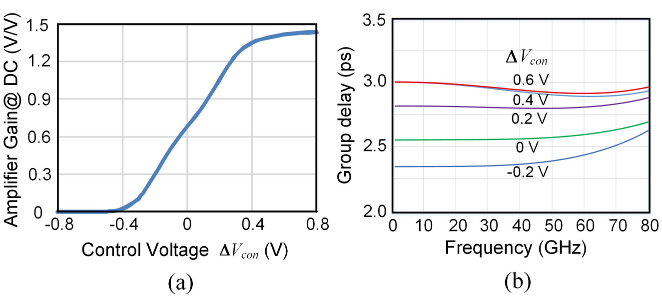


Fig. 10. Simulated (a) dc gain across ΔV_{con} and (b) group delay across frequency for different ΔV_{con} of VGA.

inductive peaking for resistor calibration over process variation and wide bandwidth.

A design procedure of the proposed n-stage delay line is summarized as follows to meet a delay tuning range of Δt_{delay}, amplitude variation of ΔV (in decibels), and dc power consumption of P_{dc}.

1) Design coupled transmission lines TL_{coupled} with a unit length (e.g., 50 μm). Choose the smallest spacing S between the two lines that the DRC spacing rule allows for the maximum coupling. Choose the linewidth W and the bottom shield metal layer to obtain $Z_{0c} = Z_{0d} = Z_0$ ($k_L = k_c$) through an EM simulation.
2) Design a VGA with a load impedance of Z_0 for a given power consumption of P_{dc}/n to achieve a bandwidth larger than the maximum design frequency.
3) Connect the VGA to TL_{coupled} with TL_{delay} to complete a single stage tunable delay line. Increase the length of TL_{coupled} and adjust the length of TL_{delay} until the delay tuning range reaches $\Delta t_{\text{delay}}/n$ with an amplitude variation $< \Delta V/n$ (in decibels) in simulation using the S-parameter model of the unit coupled lines in 1). If the condition cannot be met, increase the stage number n and return to 2).
4) After the final length of TL_{coupled} and TL_{delay} are determined, combine them in layout for a single stage and run a full EM simulation to extract S-parameters.
5) Run a full simulation of n-stage tunable delay line using the S-parameters in 4).

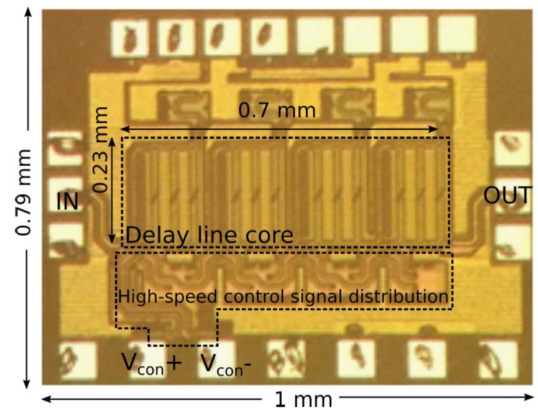


Fig. 11. Chip micrograph fabricated in a 45-nm RF SOI CMOS.

V. Measurement Results

The IC was fabricated in a 45-nm RF SOI CMOS process. The chip size is 1 mm × 0.79 mm including pads as shown in Fig. 11. The IC consumes 22 mW from 1.2-V power supply. All measurements reported in this section have been performed on-wafer. Unless otherwise indicated, measurements are taken at 25 °C.

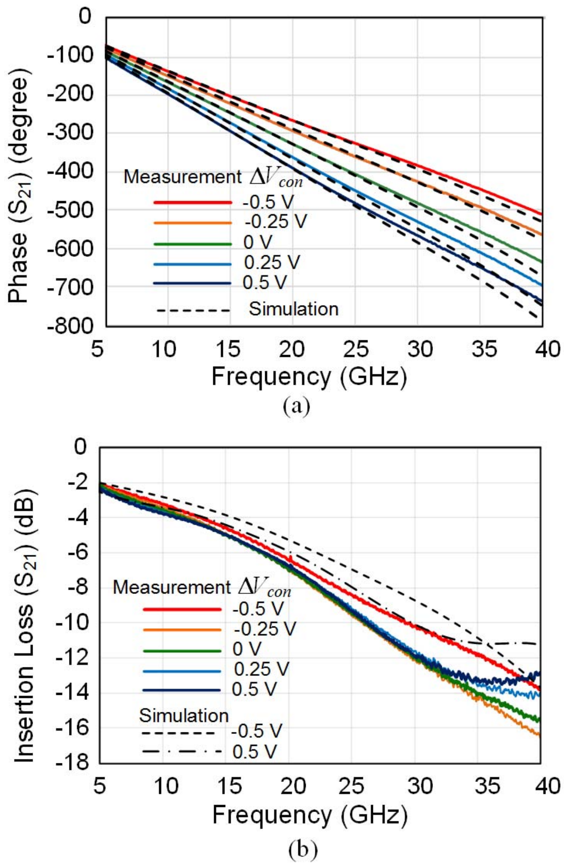


Fig. 12. Measured S_{21} in (a) phase and (b) magnitude for different ΔV_{con}.

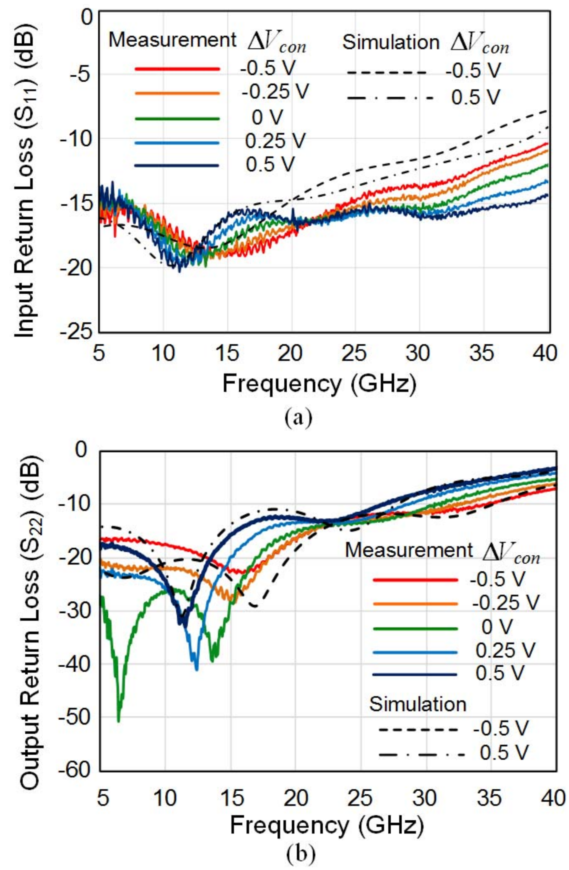


Fig. 13. Measured (a) input and (b) output return loss for different ΔV_{con}.

A. S-Parameter Measurements

S-parameters were measured using two GSG Picoprobes and an Anritsu MS4647A vector network analyzer, which was calibrated with the two-port short-open-load-through method using a CS-5 calibration substrate. Fig. 12 shows the measured S_{21}(phase and magnitude) for different control voltages from -0.5 to 0.5 V with a 0.25-V step along with simulated results. For simulation, the S-parameter model of the coupled transmission lines was extracted from the EM simulation as shown in Fig. 8, and the parasitic capacitance and resistance of the VGA were extracted using an R/C-extraction tool (Caliber [32]). More than 180° phase tuning range is achieved at 30 GHz with an average insertion loss $|S_{21}|$ of -11.6 dB and $|\Delta S_{21}| <1$ dB. $|S_{21}|$ drops around 0.3 dB/GHz over frequency below 20 GHz (available 3-dB bandwidth of 10 GHz) and drops around 0.5 dB/GHz (available 3-dB bandwidth of 6 GHz) between 20 and 30 GHz as shown in Fig. 12(b), which allows a large fractional bandwidth for carrier frequencies lower than 30 GHz. The overall insertion loss is the composite effect of different mechanisms: 1) intrinsic transmission line metal losses modeled from the EM simulation as shown in Fig. 8(c) and conductive loading from VGA inputs; 2) power transfer to the amplifier line through coupling which decreases with ΔV_{con}, as explained in (10); and 3) power transfer to the amplifier line through Miller effect with an amplifier time delay which increases with ΔV_{con}, as explained in (12a). At lower frequencies with smaller $\Delta\beta x$, the overall loss is dominated by the first mechanism, but, as frequency increases, the second and third mechanisms become the main loss contributors. Fig. 13 shows the measured input and output return losses. The input return loss does not vary significantly for different values of ΔV_{con} and is consistently below -15 dB up to 25 GHz as expected from the simultaneous change in inductance and capacitance for constant Z_0 in (3b). The output return loss is also lower than -10 dB up to 25 GHz over V_{con}. The output return loss is still below -10 dB at 30 GHz for $\Delta V_{\mathrm{con}} < 0$ and increases up to -7 dB as the VGA gain increases with $\Delta V_{\mathrm{con}} > 0$. The output return loss can be potentially improved by increasing the number of stages to more than four for the same phase tuning range, which results in a decrease of the required gain of each VGA and the length of the coupled lines per stage. For clarity of illustration, the simulated results are added only for ΔV_{con} values of -0.5 and 0.5 V in Figs. 12(b) and 13.

Fig. 14 shows the measured phase and insertion loss with respect to ΔV_{con} at frequencies of 10, 20, and 30 GHz. The phase is linearly controlled by ΔV_{con} and the loss variation over ΔV_{con} is within 0.3 dB up to 20 GHz. As the propagation phase becomes larger at higher frequencies and the power exchange between the signal and amplifier lines becomes more significant, a larger loss variation over ΔV_{con} is measured at 30 GHz. Fig. 15 shows the measured phase tuning range and insertion loss as input power increases. VGA gain compression

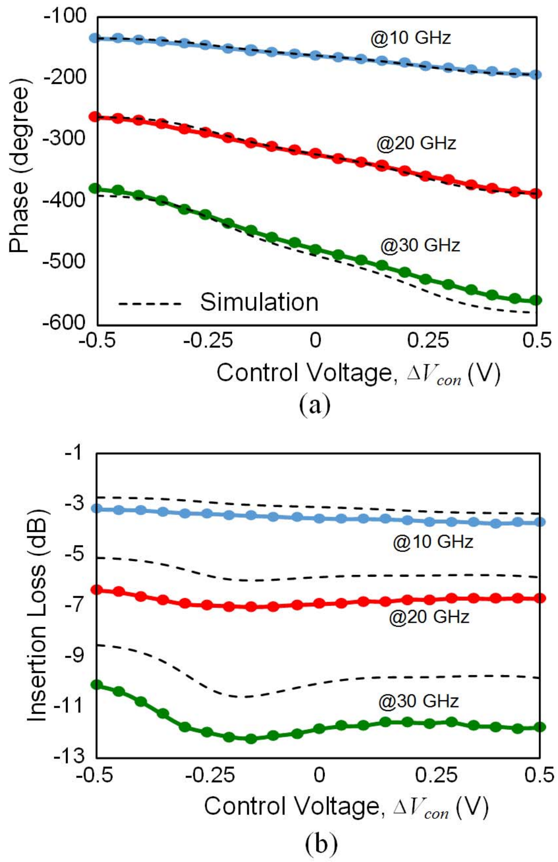


Fig. 14. Measured (a) phase and (b) insertion loss across control voltage at 10, 20, and 30 GHz.

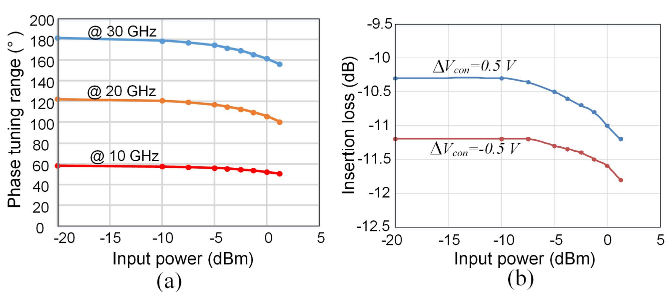


Fig. 15. Measured (a) phase tuning range and (b) insertion loss as P_{in} increases.

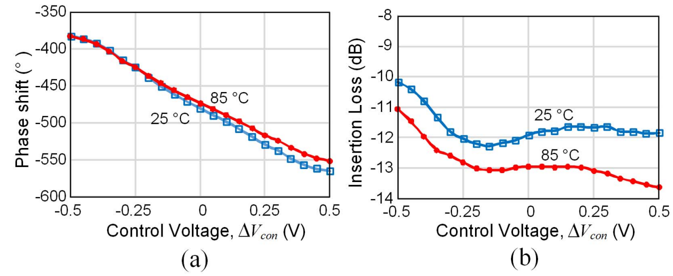


Fig. 16. Measured (a) phase shift and (b) insertion loss across control voltage ΔV_{con} at 25 and 85 °C.

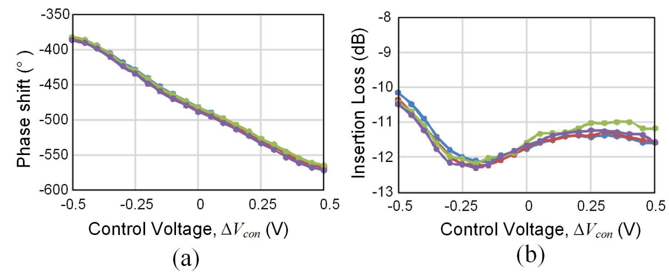


Fig. 17. Measured (a) phase shift and (b) insertion loss across control voltage ΔV_{con} among four IC samples.

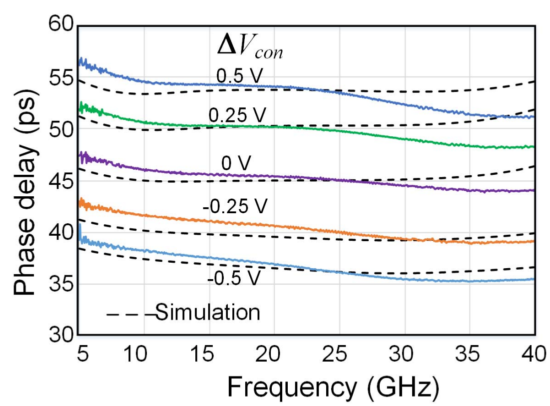


Fig. 18. Measured phase delay across frequency for different control voltages.

with higher input power compresses the tuning range by less than 10%, with less than 1-dB insertion loss degradation for $P_{\text{in}} < 0$ dBm. Fig. 16 shows the measured phase shift range and insertion loss at different temperatures. VGA gain degradation at 85 °C reduces the phase tuning range by less than 7% with insertion loss degradation less than 1.5 dB. Fig. 17 shows the measured performance among four different ICs to investigate sample-to-sample variation, which shows consistent performances among four ICs.

Phase delay ($\tau_\phi(\omega) = -(\phi(\omega)/\omega)$) and group delay ($\tau_g(\omega) = -(d\phi(\omega)/d\omega)$) are measured to demonstrate the true-time delay line feature of the design. Fig. 18 shows the measured phase delay across frequency for different values of ΔV_{con}. The measured phase delay tuning range is consistently 16.9±0.5 ps over frequency below 33 GHz. The phase delay variation between 15 and 25 GHz is less than ±2%. Fig. 19 shows the measured group delay with the group delay aperture of 10% for different values of ΔV_{con}. The group delay tuning range stays around 18±1 ps at frequencies from 11 to 24 GHz, which shows >13-GHz broadband true-time delay tunability. Both the phase delay and group delay with higher ΔV_{con} present a larger distortion at high frequencies as it is affected by the VGA gain drop.

B. Evaluation With Wideband Modulated Signals

The potential effects of the delay line's nonlinearity and group delay variation on wideband modulated signals were investigated. Toward this end, single-carrier modulated signals with 800-MHz bandwidth using 16- and 64-QAM constellations are applied to the delay line as an input signal, and the EVM of the output signal is measured as shown

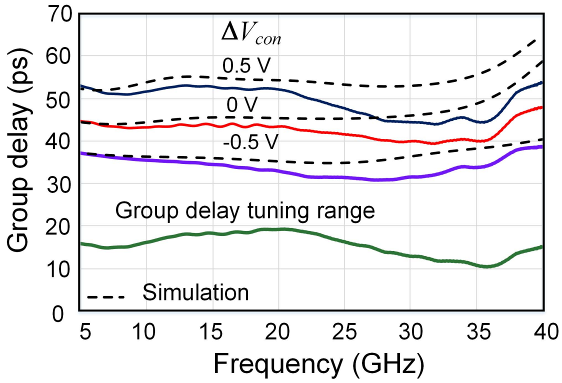


Fig. 19. Measured group delay across frequency for different control voltages and group delay tuning range.

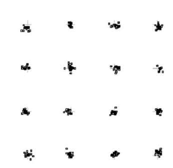
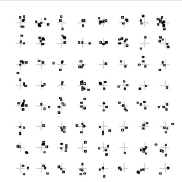
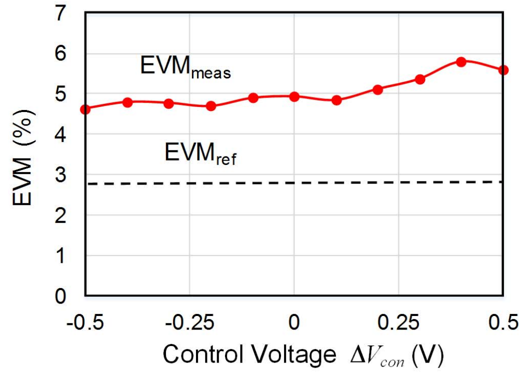
Modulation Format	16QAM	64QAM
Bandwidth	800 MHz	800 MHz
Data rates	3.2 Gbps	4.8 Gbps
Constellation		
EVM_{ref} (rms)	2.9 %	2.6 %
EVM_{meas}(rms)	4.7 %	4.3 %

Fig. 20. Measured output 16- and 64-QAM constellations and EVMs for a signal bandwidth of 800 MHz at 30 GHz.

in Fig. 20. For this experiment, the input signal is generated using a Keysight E8267D PSG vector signal generator and M8190A arbitrary waveform generator, and the output signal is received by a Keysight N9040B UXA Signal Analyzer N9040B which measures EVMs. The average power of the modulated signal applied to the delay line is approximately −5 dBm where both the loss and phase tuning range start to be compressed as shown in Fig. 15. The measured EVMs at the output of the delay line are degraded by less than 2% compared to the reference EVMs measured with an on-chip thru. The observed minor EVM degradation is mainly due to the nonlinear effects of: 1) VGA input capacitance variation with an input signal amplitude [33] and 2) VGA gain compression which affects phase response as well as amplitude response through the Miller effect. Fig. 21(a) shows the measured EVM across phase control voltage V_{con} for the 800-MHz 16 QAM signal. As ΔV_{con} increases for higher VGA gain and phase shift, the EVM increases due to a larger phase distortion. The EVM measurement for 16- and 64-QAM signals is also carried out at multiple center frequencies ranging from 29.2 to 31.6 GHz. Since 800-MHz bandwidth signals were employed, this measurement covers a continuous bandwidth of 3.2 GHz. As shown in Fig. 21(b), only a minor degradation in EVM with respect to the reference measurement is observed.



(a)

Carrier frequency	16QAM		64QAM	
	EVM_{ref}	EVM_{meas}	EVM_{ref}	EVM_{meas}
29.2 GHz	2.5 %	3.6 %	2.2 %	3.2 %
30 GHz	2.9 %	4.7 %	2.6 %	4.3 %
30.8 GHz	2.9 %	4.9 %	2.6 %	4.4 %
31.6 GHz	2.8 %	5.6 %	2.5 %	5.0 %

(b)

Fig. 21. (a) Measured EVM across delay control voltage V_{con} for 16QAM. (b) Measured EVM with 16- and 64-QAM 800-MHz modulated signals with four different carrier frequencies from 29.2 to 31.6 GHz.

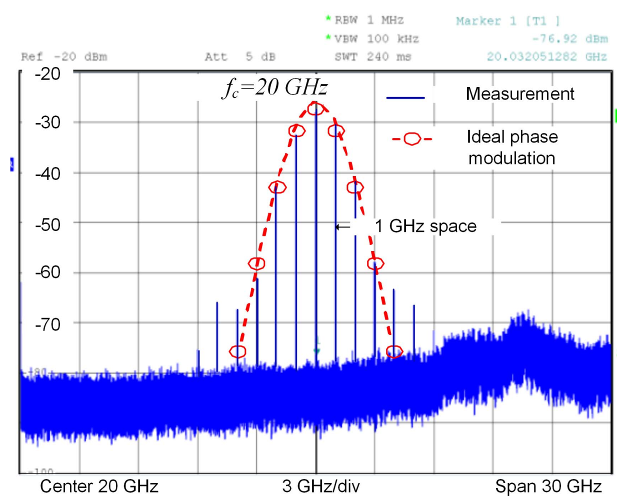


Fig. 22. Measured output spectrum at input frequency of 20 GHz with a 1-GHz phase modulation signal.

C. Measured High-Speed Phase Modulation Performance

One of the important features in the proposed structure is its high-speed phase modulation capability. Fig. 22 shows the measured spectrum resulting from modulating the proposed tunable delay line with a 3-dBm 1-GHz sinusoidal at an input frequency of 20 GHz. The input sinusoidal signal at 20 GHz is generated using an Anristu MG3697C signal generator,

TABLE I
MEASUREMENT RESULT SUMMARY AND COMPARISON TO PUBLISHED KA-BAND PHASE SHIFTERS

	This work	[5]Garg, TMTT 2017	[6]Mazor, IMS2016	[7]Shin, MWCL2016	[8]Elkholy, TMTT2018	[10]Chang, MWCL2018	[11]Akbar, TCASII2018	[25]Tousi, RFIC2016
Topology	**Distributed Miller**	Reflection type		Switched LC		Vector Modulator		Tunable transmission line
Frequency [GHz]	**30**	28	30	27.5-28.35	29	28	28	28
Phase control type	**Continuous (Linear)**	Discrete (11.25° step)	Continuous	Discrete (22.5° step)	Discrete (45° step)	Continuous	Continuous	Discrete (4.75° step)
Phase tuning range [°]	**182**	360	206	360	360	360	360	190
Time delay tuning range [ps]	**13*, 17****	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
Average Insertion loss (dB)	**11.5**	7.75	4.8	6.36	5.6	5.2	4.95	9.3
High-speed phase modulation capability	**Yes (1 GHZ)**	No	No	No	No	No	No	No
Max insertion loss variation (dB)	**1**	0.3[#]	0.4	1	0.5	1.55	0.25[##]	0.258[#]
Active area (mm^2)	**0.28[$]**	0.16	0.64	0.23	0.13	0.31	0.063[$$]	0.18
Technology	**45nm RFSOI**	65 nm CMOS	130 nm SiGe BiCMOS	120 nm SiGe BiCMOS	45nm RFSOI	180nm CMOS	130nm CMOS	130nm SiGe BiCMOS
Power consumption (mW)	**22**	0	0	0	0	6.6	27	0

*: group delay, **:phase delay.
[#]: with calibration, [##]:rms gain error with calibration.
[$]: includes a high-speed phase modulation circuitry, [$$]:excludes S2D and D2S converters.

and the output spectrum is measured using an FSU67 spectrum analyzer. The measured spectrum agrees well with the calculated ideal phase-modulation spectrum envelope with a modulation index of 1.05 derived from a full phase tuning range of ±60° at 20 GHz. This agreement demonstrates both the desired linear phase and control voltage relationship and the high-speed modulation capability of the proposed tunable delay line.

Table I summarizes the performance of the proposed design compared to the state-of-the-art Ka-band phase shifters.

VI. CONCLUSION

This paper has proposed a new type of true-time delay phase shifter exploiting the Miller effect for broadband mmWave phased array applications. This paper has presented, for the first time, a theoretical framework for understanding the Miller inductance and the associated distributed Miller effect in coupled transmission lines, and described a successful prototype IC demonstration of an mmWave phase shifter exploiting this effect. The continuous phase tuning capability and true-delay shifting features of the demonstrated design are key properties to enable both fine beam-steering resolution and wideband signal processing in beam-forming arrays for ultrawide band communication and high-resolution radar systems.

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